Operating Instructions

Gas engine-generator sets
GG06RA500A1 and GG06RA500A2

SS180185/01E

A Rolls-Royce solution

Name	SLU Name Model	Engine model	Application group
mtu 6R500 GS	GG06RA500A1	E406-ct80	3A, continuous operation, unrestricted
mtu 6R500 GS	GG06RA500A1	E406-bt80	3A, continuous operation, unrestricted
mtu 6R500 GS	GG06RA500A1	E406-ct70	3A, continuous operation, unrestricted
mtu 6R500 GS	GG06RA500A2	E406-ct80	3A, continuous operation, unrestricted
mtu 6R500 GS	GG06RA500A2	E406-bt80	3A, continuous operation, unrestricted
mtu 6R500 GS	GG06RA500A2	E406-ct70	3A, continuous operation, unrestricted

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1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper operation of your engine-generator set from the manufacturer Rolls-Royce Solutions. It is vitally important that all persons working on or with the engine-generator set are familiar with all the contents of this manual and follow the instructions provided carefully at all times.

In some cases, it may prove necessary to depart from the guidelines specified in this manual in response to extraneous circumstances. Any such departure must comply with locally applicable directives and standards. All relevant safety directives and safety precautions shall be duly considered and respected.

Following these Operating Instructions will ensure that the plant operates efficiently and reliably.

All procedural instructions and safety precautions shall be followed closely to ensure that the plant operates as intended and to avoid injury or harm. Read and follow the instructions in the "Safety" section at the beginning of this manual.

All the instructions and diagrams have been checked to ensure accuracy and user-friendliness, however the skills and qualification of the operating personnel remain the most significant factor nevertheless. Rolls-Royce Solutions cannot give any assurances for the actual results of any work performed according to the descriptions provided in this manual, nor shall Rolls-Royce Solutions be held liable for any resultant personal injury or material damage. Personnel charged with running and operating the plant do so entirely at their own risk.

Other documents included in the scope of supply shall also be given due consideration in addition to these Operating Instructions.

All documents required by Machinery Directive 2006/42/EC are included in the standard scope of documentation for the engine-generator set. The following documents are supplied in addition:
- Test records
- Spare parts catalog
- Wiring diagrams

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2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:
- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at:http://www.mtu-solutions.com

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2.2 Intended use of gas systems

Intended use

The machine is intended to be used exclusively for electrical power generation and waste heat utilization in stationary operation, either in sufficiently ventilated rooms or in container solutions.

A gas/air mixture is fed to the internal combustion reciprocating piston engine. This is converted into mechanical work in the individual cylinders through combustion. The flywheel of the engine drives the synchronous generator connected via an elastic coupling, resulting in a conversion to electric energy. The engine and the generator are joined to the base frame with resilient, vibration-damped elements.

For mains frequency 60 Hz, depending on the series either a gearbox is used between the engine and generator or the engine speed is increased accordingly. The waste heat generated during the combustion process is dissipated to the surroundings via coolers (power generator) and/or made available to the customer via heat exchangers (combined heat and power plant).

Area of application	Applicable	Comment
Industry	Yes	–
Business/Service	Yes	In some individual cases, however, such as with swimming pool facilities, schools, heating stations for apartment buildings and the like, no end user in the classic sense is involved
Household	No	Operation is excluded
Potentially explosive area	No	Operation is not permitted in potentially explosive areas
Underground	No	Operation is excluded
Mobile use	No	Operation is excluded

Table 1: Area of application of machine

There are 3 operating modes of the machine:
- Mains parallel operation
- Mains failure operation
- Isolated operation

Foreseeable misuse

Any other use, particularly misuse or wrong parametrization, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

Operation

Operation is not permissible if the physical conditions specified on the technical data sheet or on/in other documents (concerning temperature, pressure, site altitude, etc.) are not complied with. Operation is not permissible without protective equipment, usage of approved spare parts, supplied accessory parts, exclusion of conversions, warning of foreseeable misuse.

Operation is only permissible with the use of fluids and lubricants (natural gas, biogas, quality-compliant gas, quality-compliant water, lubricants) in accordance with the (→ Fluids and Lubricants Specifications).

Operation is only permissible with preservation agents approved by Rolls-Royce Solutions in accordance with the (→ Preservation and Represervation Specifications).

Operation is only permissible after proper integration/installation.

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The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:
- http://www.mtu-solutions.com

Surroundings, climate, temperature

The weather plays no role with regard to the intended use of the engine-generator set/module provided that operation takes place inside an installation room. For container applications, however, the permissible outdoor temperatures at the installation site play a decisive role. The product-specific ambient conditions such as climate and temperature are indicated in the technical description.

Prerequisite for proper use

Fulfillment of all the requirements set out in the “Installation conditions” document.

Any use that does not fall within the described limits for use is regarded as improper.

Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

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2.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:
- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.

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